

Coding & Solution

Date	3 July 2026
Team ID	SWTID-2026-8260
Project Name	A Comprehensive Measure of Well-Being using HDI Prediction System
Maximum Marks	5 Marks

Coding & Solution

Solution Summary

Field	Details
Repository Link / URL	https://github.com/nikhithad061-ux/hdi-prediction-system
Programming Language(s)	Python
Framework(s) Used	- Flask (for web app) - Pandas, NumPy - Scikit-learn - Matplotlib, Seaborn
Key Features Implemented	- HDI Prediction using ML - Data visualization - Correlation heatmap - User input-based prediction - Web interface
Pending / Incomplete Features	- Advanced UI improvements - More model optimization
Setup / Run Instructions	- Install dependencies: pip install -r requirements.txt - Run app: python app.py - Open browser: http://127.0.0.1:5000

Code Quality Checklist

S.No	Criteria	Status (Yes / No)	Remarks
1	Code is modular and organized into functions / classes	Yes	Code divided into functions for preprocessing and prediction
2	Meaningful variable and function names are used	Yes	Clear and understandable naming conventions followed
3	Code includes comments / documentation where necessary	Partial	Basic comments added, detailed documentation can be improved
4	Error handling is implemented for critical operations	Partial	Basic input validation present, advanced handling not implemented
5	The application runs without critical errors	Yes	Application runs successfully without crashes
6	Code is committed to a version control repository	Yes	Project uploaded and managed using GitHub

Additional Notes / Comments:

- The project code is well-structured and modular, ensuring readability and maintainability.
- Meaningful naming conventions are used throughout the code.
- Basic comments are included to explain key functionalities.
- The application runs successfully without critical errors.
- Version control is maintained using GitHub.
- However, advanced error handling and detailed documentation can be further improved in future enhancements.